

boundaries between the virtual and real, creating mixed reality experiences where virtual objects coexist with the real environment.

Overall, the evolution/origin of hyper-real platforms can be attributed to the ongoing advancements in technology, the exploration of immersive virtual experiences, and the desire to push the boundaries of realism in the virtual realm.

Advancements in hyper-real technologies

Hyper-real platforms are virtual environments that strive to replicate reality in a highly immersive manner. These platforms utilise advanced technologies such as virtual reality (VR), augmented reality (AR), haptic feedback, photorealistic graphics, and artificial intelligence (AI) to create a sense of presence and realism. By integrating these technologies, hyper-real platforms offer users a heightened level of immersion and interaction, blurring the boundaries between the physical and virtual worlds.

Virtual reality (VR) provides users with a fully immersive experience, transporting them to digital environments, while augmented reality (AR) overlays virtual elements onto the real world, enhancing users' perceptions and interactions. Haptic feedback technologies enable users to feel physical sensations in the virtual environment. Sensory immersion includes techniques like smell and temperature simulation, further enhancing realism. Real-time ray tracing and advanced rendering techniques allow hyper-real platforms to achieve stunning visual fidelity.

High-resolution textures, realistic lighting, and lifelike animations contribute to a more authentic experience. AI algorithms enable dynamic and intelligent interactions within hyper-real platforms. Machine learning techniques improve user interactions, object recognition, and natural language

processing, enhancing overall realism. Hyper-real platforms revolutionise gaming experiences, providing realistic simulations, immersive storytelling, and interactive gameplay. Users can engage in virtual sports, explore fantastical worlds, and connect with other players in hyper-realistic environments. Hyper-real platforms offer enhanced virtual conferencing capabilities, enabling lifelike interactions, spatial audio, and realistic environments.

Remote collaboration becomes more immersive, fostering better communication and engagement among participants. Industries such as aviation, healthcare, and defence utilise hyper-real platforms for training simulations. Realistic scenarios enable trainees to practice skills, make critical decisions, and experience lifelike consequences in a safe environment.

Architects and designers leverage hyper-real platforms to create virtual walkthroughs and visualisations of buildings and spaces. Clients can experience and provide feedback on architectural projects before they are physically constructed. Hyper-real platforms facilitate virtual therapy sessions, pain management, and phobia treatment through immersive experiences.

Patients can engage in simulations that expose them to controlled environments, enabling gradual exposure and therapeutic interventions. Hyper-real platforms raise concerns about data privacy, consent, and potential misuse of realistic virtual scenarios.

Ensuring user safety, consent, and responsible use of personal information is crucial. Technical limitations and infrastructure requirements play a vital role in hyper-real platform design. Achieving hyper-realistic experiences requires powerful hardware and infrastructure, which may limit accessibility for some users. Bandwidth requirements, latency issues, and device compatibility pose challenges for widespread adoption. Creating hyper-realistic content requires significant

resources, expertise, and time. User-generated content platforms and tools need to be developed to empower users to contribute to the metaverse.

Hyper-real platforms should prioritise accessibility features to cater to users with disabilities. Ensuring inclusivity and designing interfaces that accommodate diverse user needs is essential. These platforms have the potential to redefine social interactions, allowing people to connect and communicate in new ways. Relationships may be formed and nurtured within virtual environments, challenging traditional notions of interpersonal connections.

Hyper-real platforms create new avenues for businesses, enabling virtual commerce, advertising, and brand experiences. The emergence of virtual economies and digital assets within the metaverse presents economic opportunities for content creators and entrepreneurs. Collaboration among hyper-real platform developers is crucial to establish standards, interoperability, and shared experiences.

Open source initiatives and partnerships can drive innovation and accelerate the development of the metaverse. As the metaverse continues to develop, hyper-real platforms will shape the way we interact, work, play, and experience the digital world.

Use cases of hyper-real platforms in metaverse

From gaming and entertainment to architecture, education, retail, healthcare, and beyond, the transformative use cases of hyper-real platforms are vast and diverse (Table 1). As technology advances and these platforms continue to evolve, we can anticipate even more groundbreaking applications that will shape the future of the metaverse, unlocking new levels of realism and interaction that were once unimaginable. Some of the popular use cases where hyper-real platforms can play a vital role are listed below.